

Project Documentation: SmartCare CRM (Healthcare Management System)

- **Author:** Chinta Gayathri Sreeja
 - **System:** Salesforce Lightning Platform
-

1. Executive Summary

1.1. The Problem:

The hospital's original process for managing patients and appointments relied heavily on manual paperwork, phone calls, and scattered spreadsheets. This caused frequent scheduling conflicts, missing patient records, and limited visibility for doctors and administrators. Patient experience suffered due to delays and lack of proper follow-ups.

1.2. The Solution:

A centralized Salesforce application, **SmartCare CRM**, was developed to automate patient registration, appointment scheduling, doctor assignment, and prescription management. The system enforces data integrity through validation, automates reminders via flows, and provides dashboards for better insights.

1.3. The Business Value:

SmartCare CRM enables hospitals to deliver faster, more reliable care. Doctors are notified automatically of their appointments, patients receive reminders, and administrators gain visibility into hospital operations. This improves efficiency, reduces manual errors, and enhances overall patient satisfaction.

2. Org Setup & Configuration

2.1. Company Profile: The hospital's default time zone and currency were set to ensure appointment timings and billing data are accurate.

2.2. Business Hours & Holidays: A custom **Hospital Working Hours** calendar was created, along with official holidays, to accurately manage scheduling and availability for appointments.

2.3. User Creation: Users were created for the key roles:

- **Hospital Admin** (System Administrator)
- **Doctor** (Medical Practitioner using the system)
- **Receptionist** (Staff handling patient registrations and scheduling)

New User Reset Password(s) Add Multiple Users						
☐ Action	Full Name ↑	Alias	Username	Role	Active	Profile
☐ Edit	Admin_Hospital	admin	admin@smartcare.com		✓	System Administrator
☐ Edit Login	Anjali_Rao_Dr.	Anjali	dranjali@smartcare.com	Doctor	✓	Doctor_Profile
☐ Edit	Chatter_Expert	Chatter	chatty.00d9k000007a1ehuaq.ro2xn1dfo2@chatter.salesforce.com		✓	Chatter_Free_User
☐ Edit Login	EPIC_OrgFarm	QEPIC	epic.1b899cc09249@orgfarm.salesforce.com		✓	System Administrator
☐ Edit Login	Riya_Receptionist	Riya	riyatec@smartcare.com	Receptionist	✓	Receptionist_Profile
☐ Edit	User_Integration	integ	integration@00d9k000007a1ehuaq.com		✓	Analytics.Cloud.Integration.User
☐ Edit	User_Security	sec	insightssecurity@00d9k000007a1ehuaq.com		✓	Analytics.Cloud.Security.User
New User Reset Password(s) Add Multiple Users						

2.4. Custom Profiles & Permission Sets:

- Profiles:** Hospital Admin, Doctor, Receptionist — each configured with access to the relevant objects.

New Profile		A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Other All																									
☐ Action	Profile Name ↑																										
☐ Edit Clone	Contract Manager																										
☐ Edit Del...	Custom_Marketing_Profile																										
☐ Edit Del...	Custom_Sales_Profile																										
☐ Edit Del...	Custom_Support_Profile																										
☐ Edit Del...	Doctor_Profile																										
☐ Edit Clone	Marketing User																										
☐ Edit Clone	Minimum Access - Salesforce																										
☐ Edit Del...	Read Only																										
☐ Edit Del...	Receptionist_Profile																										
☐ Edit Clone	Solution Manager																										
☐ Edit Clone	Standard User																										
☐ Edit Clone	System Administrator																										
☐ Edit Del...	TCS profile																										
☐ Edit Del...	Test profile																										

- Permission Sets:** Extra permissions were granted for specific tasks, such as managing sensitive patient records or updating appointment statuses.

3. Data Model & Relationships

The data model was designed to represent patients, doctors, and their medical interactions.

3.1. Key Objects Created:

- Patient (Custom Object):** Stores patient demographics and medical history.

Patient				
Fields & Relationships				
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Blood Group	Blood_Group__c	Picklist		
Contact Number	Contact_Number__c	Phone		
Created By	CreatedById	Lookup(User)		
Date of Birth	Date_of_Birth__c	Date		
Email	Email__c	Email		
Emergency Contact	Emergency_Contact__c	Text(20)		
Gender	Gender__c	Picklist		
Last Modified By	LastModifiedById	Lookup(User)		
Medical History	Medical_History__c	Long Text Area(5000)		
Owner	OwnerId	Lookup(User,Group)		✓

- **Doctor (Custom Object):** Stores doctor details, including specialization.

SETUP > OBJECT MANAGER

Doctor

Details

Fields & Relationships
10 Items. Sorted by Field Label

Q Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Availability End	Availability_End__c	Time		
Availability Start	Availability_Start__c	Time		
Consultation Fee	Consultation_Fee__c	Currency(5, 0)		
Contact Number	Contact_Number__c	Phone		
Created By	CreatedById	Lookup(User)		
Doctor Name	Name	Text(80)		✓
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Specialization	Specialization__c	Picklist		

Obtain Access

- **Appointment (Custom Object):** Tracks appointment date, time, status, assigned doctor, and patient.

SETUP > OBJECT MANAGER

Appointment

Details

Fields & Relationships
13 Items. Sorted by Field Label

Q Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Appointment DateTime	Appointment_DateTime__c	Date/Time		
Appointment Number	Name	Auto Number		✓
Assigned Doctor	Assigned_Doctor__c	Lookup(Doctor)		✓
Consultation Fee	Consultation_Fee__c	Currency(4, 0)		
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Patient	Patient__c	Lookup(Patient)		✓
Reason	Reason__c	Long Text Area(32768)		
Record Type	RecordTypeId	Record Type		✓
Reminder Sent	Reminder_Sent__c	Checkbox		

Obtain Access

- **Prescription (Custom Object):** Linked to appointments to track prescribed medicines and advice.

SETUP > OBJECT MANAGER

Prescription

Details

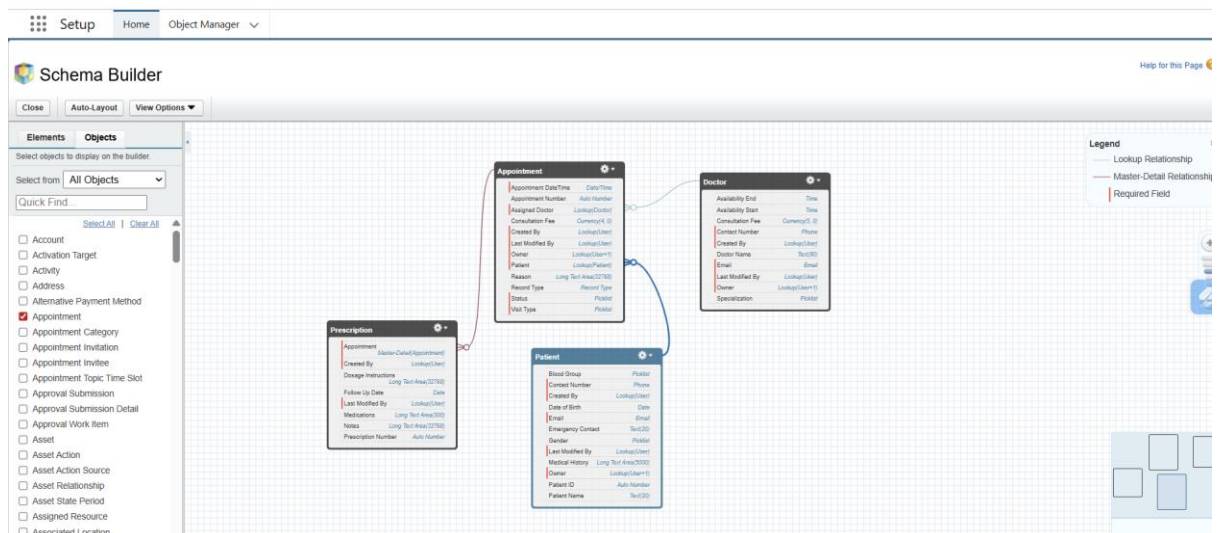
Fields & Relationships
8 Items. Sorted by Field Label

Q Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Appointment	Appointment__c	Master-Detail(Appointment)		✓
Created By	CreatedById	Lookup(User)		
Dosage Instructions	Dosage_Instructions__c	Long Text Area(32768)		
Follow Up Date	Follow_Up_Date__c	Date		
Last Modified By	LastModifiedById	Lookup(User)		
Medications	Medications__c	Long Text Area(300)		
Notes	Notes__c	Long Text Area(32768)		
Prescription Number	Name	Auto Number		✓

3.2. Relationships:

- Appointment → Patient (Lookup)
- Appointment → Doctor (Lookup)
- Prescription → Appointment (Master-Detail)



3.3. Record Types: Two record types were defined for appointments:

- New Consultation
- Follow-Up

Appointment				
2 Items. Sorted by Record Type Label				
RECORD TYPE LABEL	DESCRIPTION	ACTIVE	MODIFIED BY	
Follow-Up		✓	Hospital Admin, 9/23/2025, 6:06 AM	▼
New Consultation		✓	Hospital Admin, 9/23/2025, 5:59 AM	▼

4. Process Automation

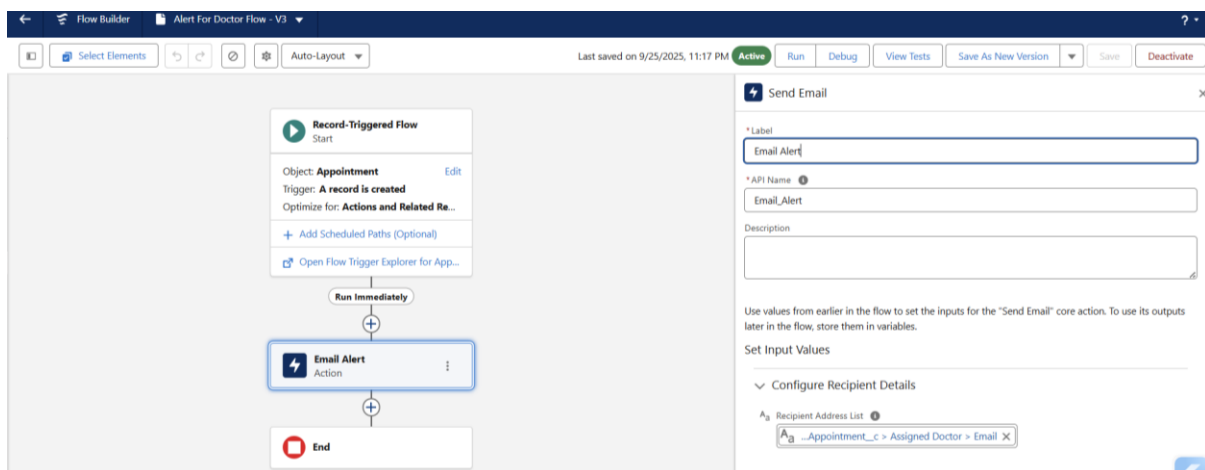
Automation was implemented to ensure smooth hospital operations.

4.1. Validation Rule: An appointment cannot be saved without a **Reason for Visit**, ensuring completeness of patient records.

The screenshot shows the 'Appointment Validation Rule' configuration page. The left sidebar contains navigation links: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, and Field Sets. The main content area is titled 'Appointment Validation Rule' and includes a 'Back to Appointment' link. Below the title is the 'Validation Rule Detail' section with fields for Rule Name, Error Condition Formula, Error Message, Description, Created By, and Modified By. The rule is named 'Appointment_Not_in_Past', the formula is 'Appointment_DateTime__c < NOW()', the error message is 'Appointment date/time cannot be in the past', and the error location is 'Appointment DateTime'. The rule is active and was created by 'Hospital Admin' on 9/25/2025 at 3:10 AM.

4.2. Flows:

- **Doctor Notification Flow:** Notifies doctors automatically when they are assigned to a new appointment.



4.3. Approval Process : Could be extended in future for requests such as critical treatment approvals.

The screenshot shows the 'Approval Processes' configuration page for 'Appointment Approval'. The left sidebar contains navigation links: Data, Mass Transfer Approval Requests, Feature Settings, Approval Settings, and Process Automation. The main content area is titled 'Approval Processes' and includes a 'Back to Approval Process List' link. Below the title is the 'Process Definition Detail' section with fields for Process Name, Unique Name, Description, Entry Criteria, Record Editability, Approval Assignment Email Template, Initial Submitters, Created By, and Modified By. The process is named 'Appointment Approval', the unique name is 'Appointment_Approval', the description is 'Approval for an Appointment', the entry criteria is 'Appointment__Status EQUALS Requested', the record editability is 'Administrator OR Current Approver', the approval assignment email template is 'New Appointment Notification', the initial submitters are 'Appointment Owner, Group Doctors, Role: Doctor, User: Dr. Anjali Rao', and the process is active. Below the process definition detail is the 'Initial Submission Actions' section, which includes an action 'Record Lock' with the description 'Lock the record from being edited'. The 'Approval Steps' section shows a single step 'Doctor Approval' with the assigned approver 'User: Dr. Anjali Rao' and the reject behavior 'Final Rejection'. The 'Final Approval Actions' section is also visible.

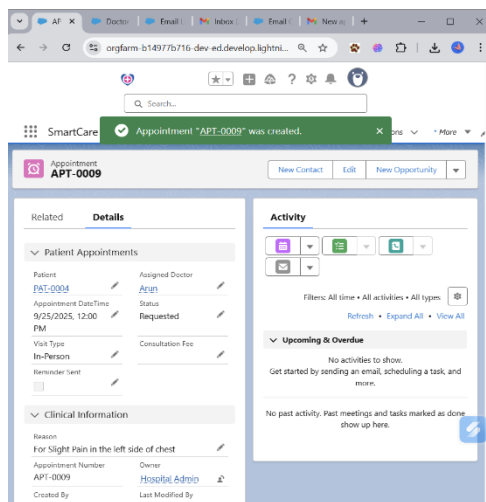
5. Apex Programming

5.1. Trigger: A trigger was created on **Appointment__c** to ensure that each appointment record includes a reason before insertion.

```
AppointmentTrigger.apxt
Code Coverage: None API Version: 64 Go To

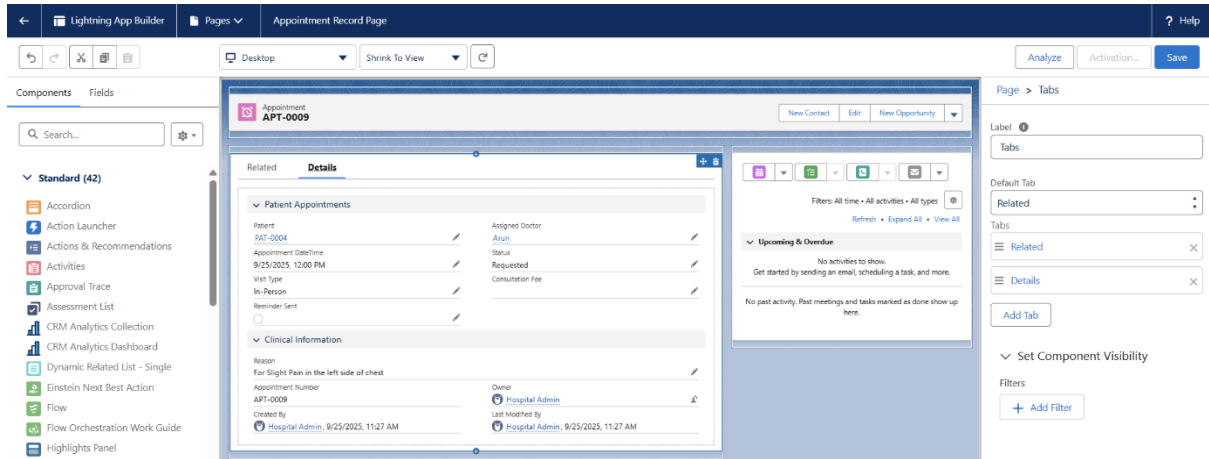
1 trigger AppointmentTrigger on Appointment__c (before insert) {
2   for (Appointment__c appt : Trigger.new) {
3     if (String.isEmpty(appt.Reason__c)) {
4       appt.addError('Reason is required for the appointment.');
```

5.2. Testing: Both trigger and class were tested successfully by creating sample records in the Developer Console.

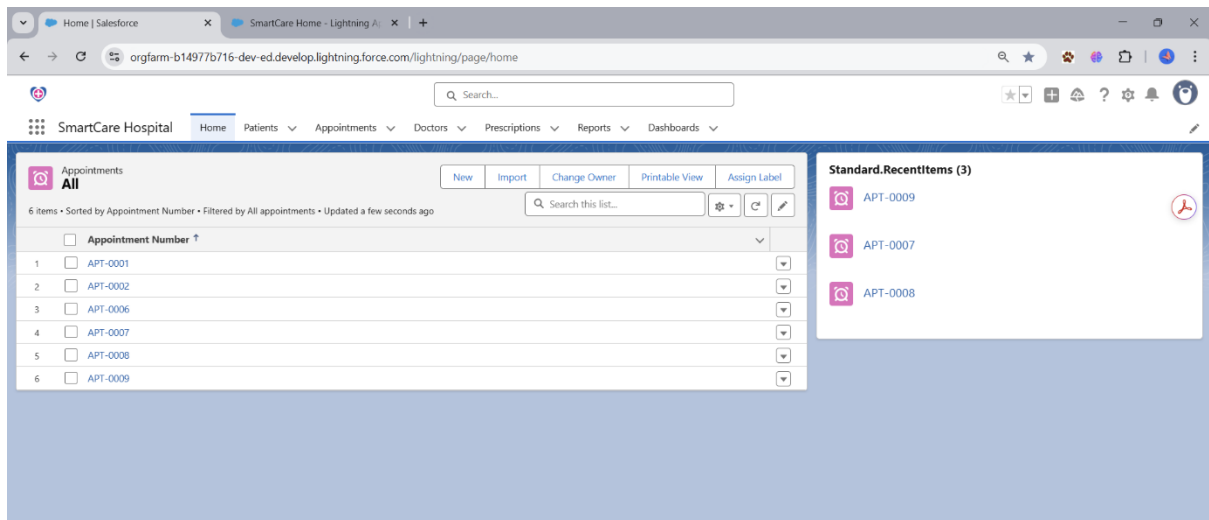


6. User Interface Development

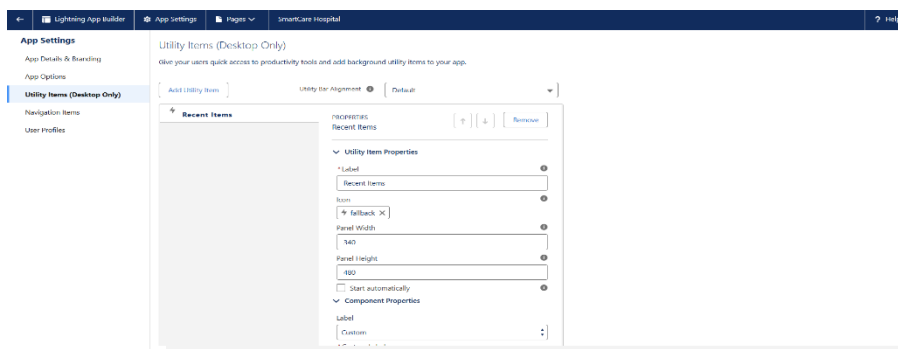
6.1. Custom Record Pages: Appointment record pages were customized using **Lightning App Builder**. Tabs were added to separate details from related lists, improving readability.



6.2. Custom Home Page: A SmartCare CRM App Home Page was designed to display a list of upcoming appointments and pending patient check-ins, acting as a quick-access dashboard for receptionists and doctors.

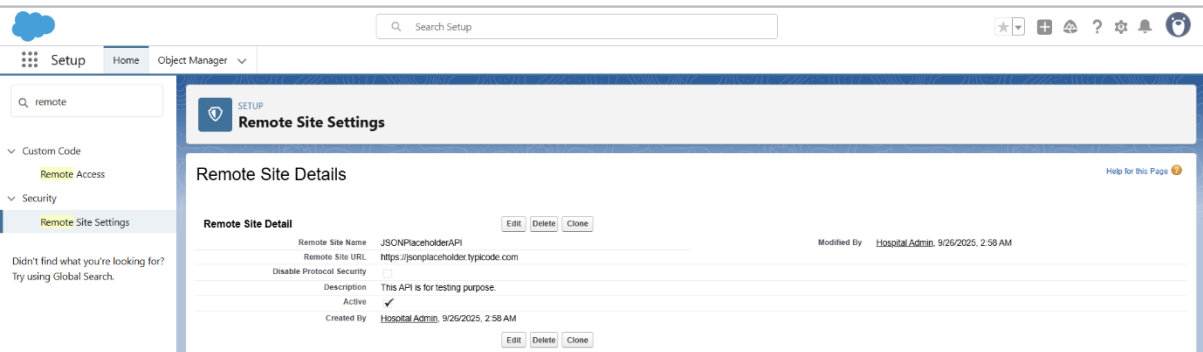


6.3. Utility Bar: A Utility Bar was added with Recent Items for quick navigation, improving the user experience.

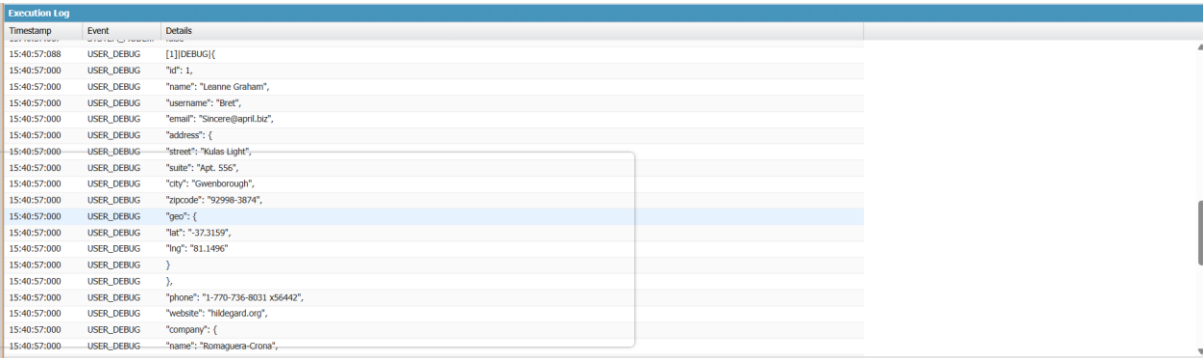
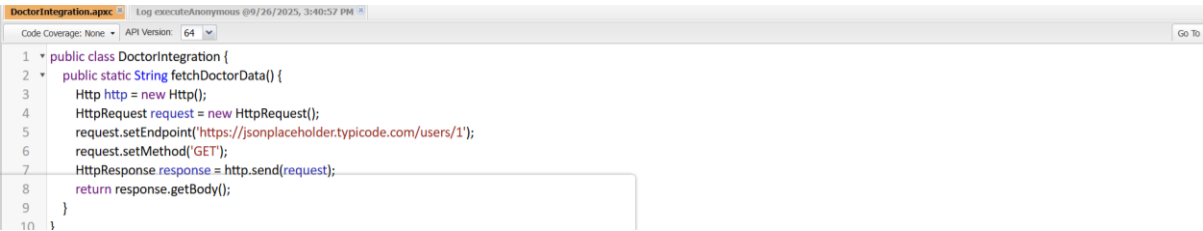


7. Integration & External Access

7.1. Remote Site Settings: Configured to allow Salesforce to connect to external APIs.

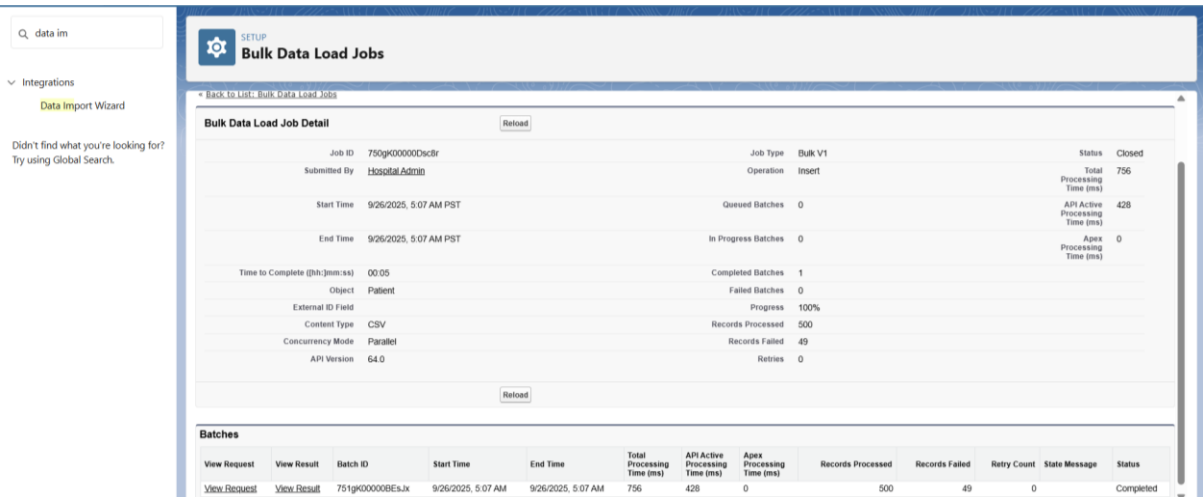
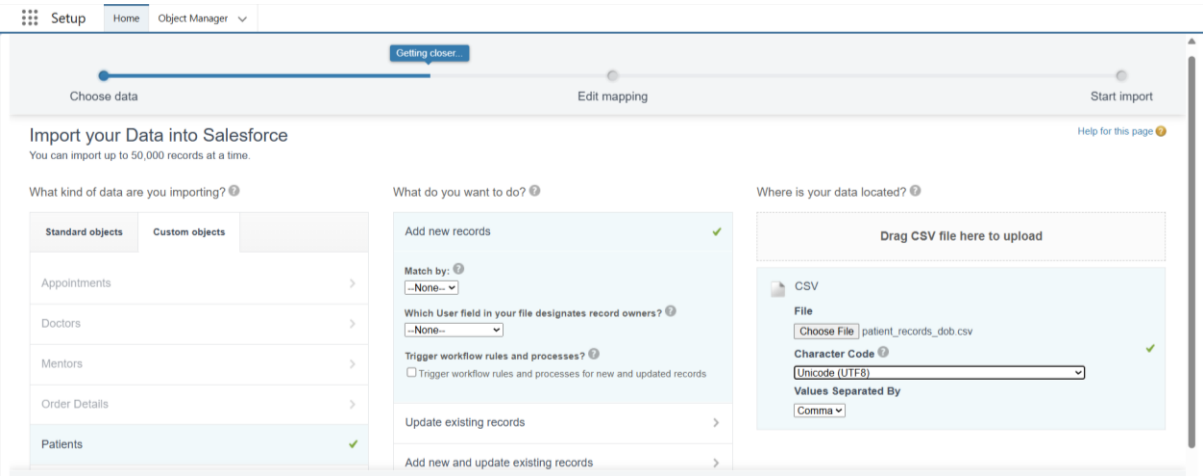


7.2. Example Integration: JSONPlaceholder API was integrated to demonstrate how external data can be fetched into Salesforce securely. This laid the foundation for potential future integrations, such as connecting with lab systems or insurance portals.

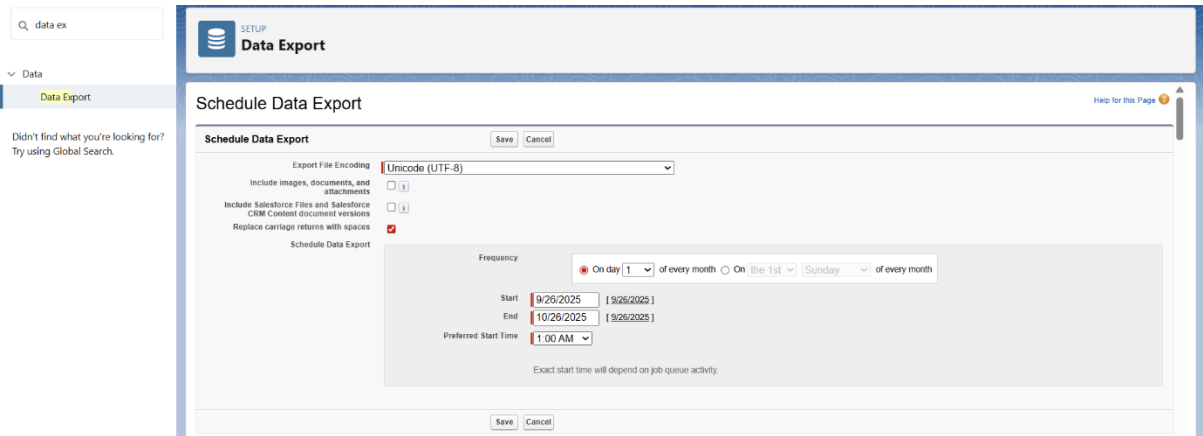


8. Data Management & Deployment

8.1. Data Import: Using the **Data Import Wizard**, 500 Patient records were imported from a prepared CSV file, ensuring bulk data entry was fast and accurate.



8.2. Data Backup: A weekly **Data Export** schedule was set up, ensuring that all patient, appointment, and doctor records are backed up regularly for recovery and compliance.



9. Reporting, Dashboards & Security Review

9.1. Reports & Dashboards: Custom reports and dashboards were created to track hospital performance metrics, such as:

- Appointments for Assigned Doctors.

Report: Appointments

Appointments Report

Enable Field Editing

Q

Add Chart

▼

🔗

Edit

▼

Total Records

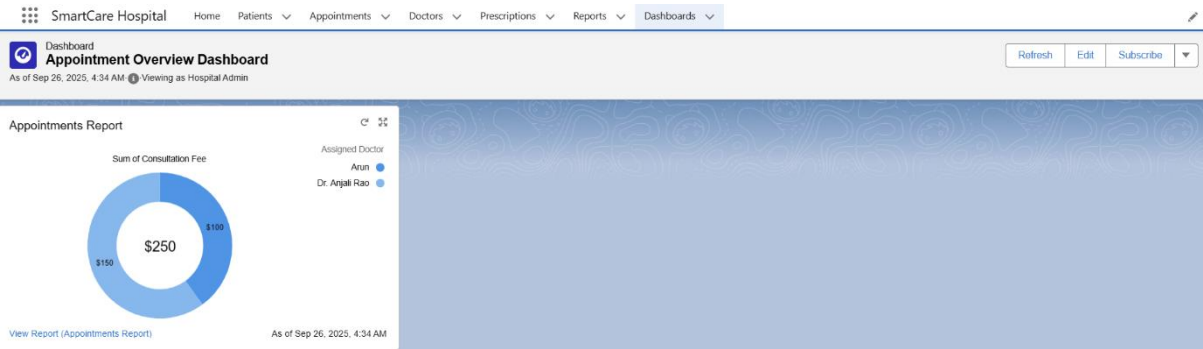
5

Total Consultation Fee

\$250

Assigned Doctor	Appointment: Appointment Number	Appointment Date Time	Status	Visit Type	Reason	Patient	Consultation Fee
Anun (2)	APT-0001	9/25/2025, 12:00 PM	Requested	In-Person	-	PAT-0002	\$100
	APT-0009	9/25/2025, 12:00 PM	Requested	In-Person	For Slight Pain in the left side of chest	PAT-0004	-
Subtotal							\$100
Dr. Arjati Rao (3)	APT-0006	9/25/2025, 12:00 PM	Requested	In-Person	-	PAT-0002	-
	APT-0008	9/25/2025, 8:00 PM	Requested	In-Person	-	PAT-0004	-
	APT-0010	9/25/2025, 12:30 PM	Requested	Teleconsultation	Head ache is not reduced	PAT-0004	\$150
Subtotal							\$150
Total (5)							\$250

- Dashboard includes the total consultation fee each doctor get.



A central dashboard was made available for administrators to monitor hospital performance in real time.

9.2. Security Review: Profiles and permission sets were reviewed to ensure the **least privilege principle**. Doctors had access only to their assigned appointments and patients, while Admins retained full control. Audit Trail was checked to track configuration changes.

